

Unequal Job Opportunities and the Measurement of Well-Being Inequality

Seminar working paper — first complete discussion draft

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Abstract

Random Utility–Random Opportunity (RURO) is a job-choice framework in which preferences rank employment packages while a separate household-specific opportunity density governs their availability. We ask how much inequality in money-metric well-being is associated with unequal job access and earning opportunities rather than heterogeneous preferences, accounting separately for non-labour resources and household composition. The empirical applications are single adults and couples in France. The incremental contribution is to carry an occupation-augmented RURO structure through tax-benefit pricing, ex-ante money-metric welfare, and grouped Shapley attribution, then compare it with a re-estimated common-opportunity model. This is not a claim to invent joint preference–opportunity estimation. Current estimates were produced by the sampled-alternatives criterion and unbounded wage density presently implemented. A corrected estimator, bounded and renormalised wage support, and affected downstream results are pending; consequently the paper reports those estimates as provisional evidence and draws no conclusion that requires corrected magnitudes.

Standing result-status note. Verified facts include the source-to-sample reconciliation, the formulas executed by the historical pipeline, and the current chosen-row-free common welfare support. Every numerical behavioural or welfare result below is a current-implementation estimate, not a corrected bounded-support estimate. The estimator, wage support, parameters, predictions, welfare levels, and decomposition results are under correction. The corrected resource classification also makes the earlier nested resource/composition percentages historical only, so those percentages are withheld. This status follows the target-model and integrability audit, the sampled-alternatives criterion audit as qualified by the welfare mission, and the source-to-estimation-sample audit. WINT-1 establishes that the welfare integrator itself is unaffected by the chosen-row term: its common support excludes the chosen row, whose contribution is exactly zero.

1 Introduction

Observed labour supply combines what a household wants with what it can reach. A worker may choose short hours because time is valuable, because longer-hour jobs are scarce, or because the available long-hour jobs pay poorly after taxes and transfers. A non-worker may have a strong preference for work but face a thin opportunity distribution. Treating all observed differences as tastes would misdescribe the economic mechanism; treating every non-preference circumstance as a job opportunity would be equally misleading. Our exhaustive accounting therefore distinguishes preferences, job access, earning opportunities, non-labour resources, and household composition or needs.

RURO separates two surfaces. Deterministic utility assigns value to consumption and leisure. A structural opportunity density assigns relative availability to employment, hours, occupations,

and wages. Extreme-value shocks then generate probabilistic choice from their product. The distinction matters for welfare because ex-ante well-being depends on the distribution of potential jobs, not only on the chosen outcome. It also matters for interpretation: estimated opportunity densities are neither literal counts of accessible jobs nor recovered deterministic capability sets.

1.1 Four strands of literature and the incremental contribution

The first strand models labour supply as choice among latent jobs. The Aaberge–Dagsvik–Strøm tradition combines nonlinear tax schedules with random job opportunities, and later work studies representations of restricted choice sets (Aaberge et al., 1995, 1999, 2009). Dagsvik and Strøm develop a sectoral latent-job formulation, while Dagsvik and Jia make the identification problem created by unobserved heterogeneity explicit (Dagsvik and Strøm, 2006; Dagsvik et al., 2014; Dagsvik and Jia, 2016). Capéau, Decoster and Dekkers provide the closest RURO estimation-and-simulation application (Capéau et al., 2016). The present paper inherits joint preference–opportunity estimation from this literature; its novelty is not the RURO label.

The second strand asks how welfare comparisons can respect heterogeneous preferences. Equivalent-income methods use a declared reference situation to map household-specific utility into a common money unit (Dagsvik and Karlström, 2005; Decoster and Haan, 2015; Decancq et al., 2015; Bargain et al., 2013). Our cross-sectional object is a level of ex-ante money-metric well-being under a declared flat-consumption reference, not a reform-specific compensating variation.

The third strand connects latent jobs to responsibility-sensitive welfare. Jacquet et al. (2026) study a reform comparison for joint-choice couples and ask how responsibility assignments affect fairness measurement. Our question is different: cross-sectional inequality in well-being, with singles and couples as parallel core applications, under one fixed policy system. We do not claim to implement the full family of measures developed in the separate Haydar–Maniquet theory project.

The fourth strand attributes nonlinear distributional statistics to inputs. The Shapley approach averages marginal contributions over orderings (Shorrocks, 2013); grouped Shapley–Owen methods preserve a declared hierarchy among factors (Audoly et al., 2025). Here the hierarchy first separates preferences from all non-preference circumstances, then separates access, earnings, resources, and composition. Algebraic exhaustiveness does not convert that allocation into causal identification or a moral-responsibility theorem.

The exact empirical increment is therefore an integrated architecture: an occupation-augmented RURO model for single and joint couple choices; detailed tax-benefit pricing; an opportunity-sensitive, preference-respecting money metric; a grouped distributional attribution; and a re-estimated common-opportunity comparison. The closest pieces exist separately in the literature. Their disciplined integration, together with explicit accounting for what the data do not identify, is the contribution.

2 Data and institutional setting

The delivered microdata are the French EUROMOD input file `FR_2016_a3`. Collection occurred in 2016, while the income reference year is 2015. The production connector binds that delivery to the `FR_2015` policy system. Amounts entering utility are monthly euros on the delivery’s price scale; wages are euros per hour and labour supply is weekly hours. Counterfactual monthly gross earnings split ordinary and overtime inputs,

$$Y^{reg}(w, h) = \frac{52}{12}w \min\{h, 35\}, \quad Y^{ot}(w, h) = \frac{52}{12}w \max\{h - 35, 0\}. \quad (1)$$

Here w is hourly pay and h is weekly hours. This split is required because equal total earnings with different hours need not be identical policy inputs. The budget map is

$$C_i(j) = \max\left\{1, \mathcal{B}_i^{FR-2015}(Y^{reg}(j), Y^{ot}(j), h(j); r_i, d_i)\right\}, \quad (2)$$

where r_i contains non-labour resources and housing inputs, d_i contains composition and policy-relevant characteristics, and $C_i(j)$ is disposable consumption in euros per month after the one-time take-up adjustment. The one euro floor is a positive-domain modelling device, not a subsistence estimate or a correction to raw data.

The corrected source audit establishes the EUROMOD connector version, model root, policy-file digest, and a field-level dictionary for all delivered inputs. It also establishes that annual employee cash earnings equal twelve times the calendar-year-average monthly equivalent; multiplying that monthly equivalent by months worked is incorrect. The hourly wage variable is a predicted or imputed construction. Positive-income identities hold for most workers, but some employees have zero recorded employee earnings, so the paper does not call every wage observed.

2.1 Parallel samples

The analysis contains 1,555 single-adult households and 2,275 opposite-sex couple households. Both arise from one screen sequence: household composition; all decision makers aged 20–60; no current education; no relevant pension status; labour-force status restricted to employee, unemployed, or inactive; no additional employable or materially earning member; and the final hours and wage restrictions. These restrictions define the target population. They are not statements that other households are behaviourally inflexible.

The former ten-hour lift was a processed-data error. In the corrected chosen rows, minimum positive weekly hours are 6 for singles, 9 for couple men, and 6 for couple women. Values at the seventy-hour cap combine recorded values and projections from above the cap. Low-hour employees and capped observations therefore require an explicit observation/support decision before final regeneration.

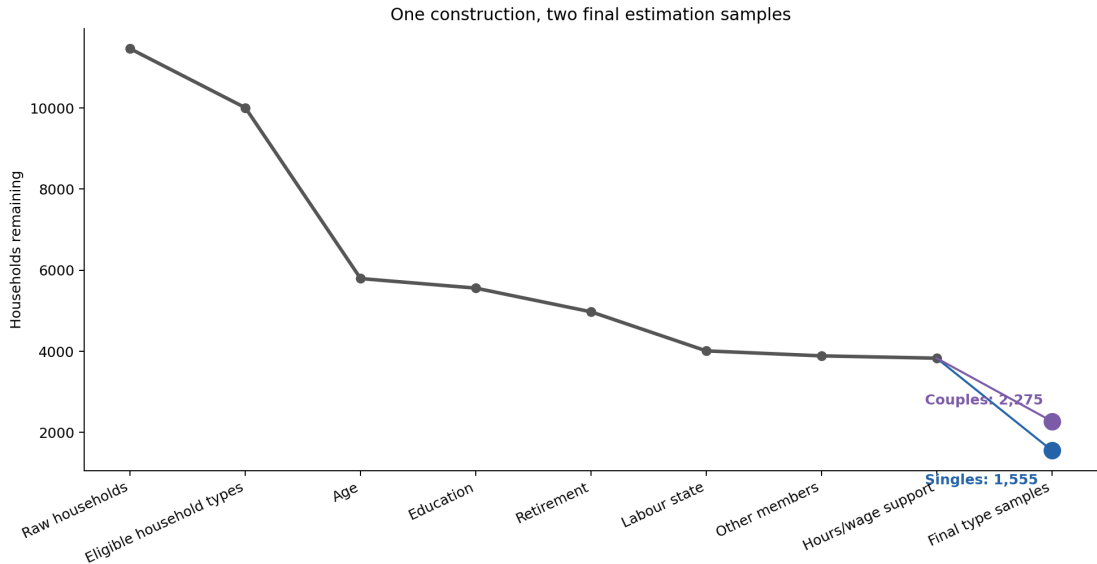


Figure 1: Population: delivered French households screened into the final single-adult and both-flexible couple samples. Units: unweighted household counts. Measure: observed sample construction. Reference: 2016 delivery, 2015 income year and FR_2015 policy system. Status: regenerated descriptive evidence; no model output.

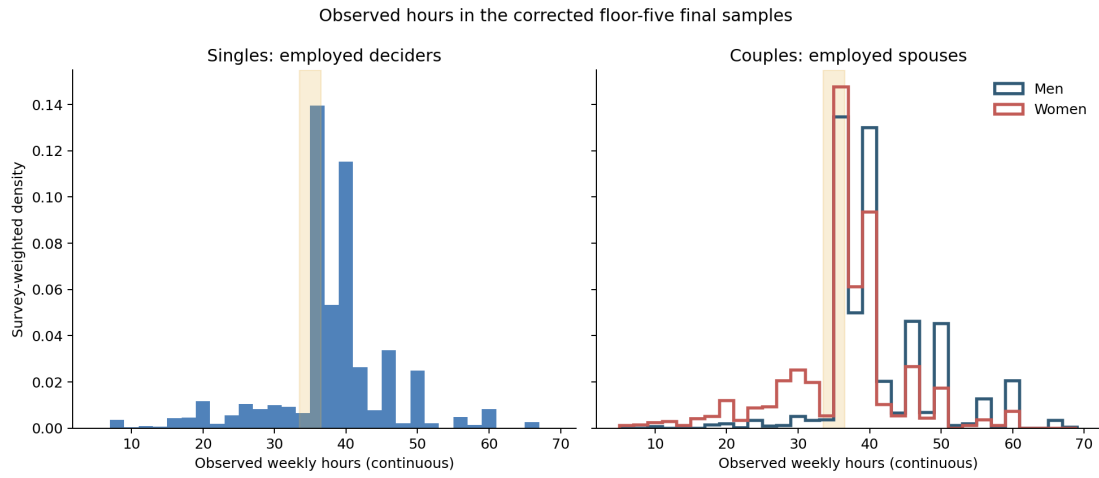


Figure 2: Population: employed decision makers in the final single and couple samples. Units: weekly hours. Measure: survey-weighted observed chosen-row distribution; named structural bands are overlaid for orientation. Reference: own observed status and hours. Status: regenerated descriptive evidence, not a model-implied distribution and not evidence of point masses.

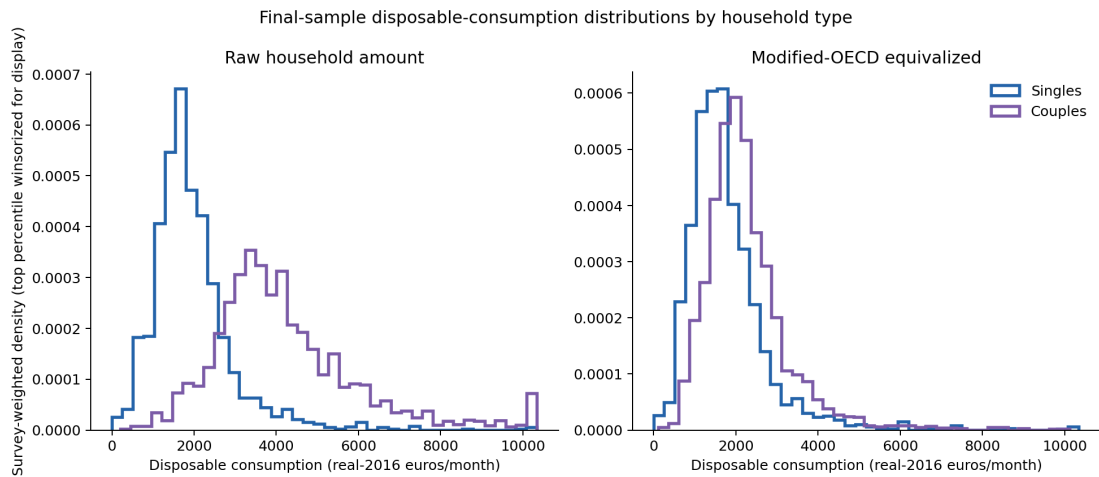


Figure 3: Population: final single-adult and couple households. Units: monthly euros, raw and modified-OECD-equivalized as identified in the panels. Measure: survey-weighted observed chosen-row disposable-consumption distribution. Reference: actual household composition. Status: regenerated descriptive evidence; no behavioural or welfare estimate.

Calculation, rationale, assumptions, alternative. The data stage calculates a decision unit, observed package, policy-priced budget inputs, and survey-representative descriptives. It is used because joint choices and nonlinear taxes must be represented at the household level. It assumes the stated timing, roster links, target-population restrictions, and one-time take-up imputation. Dropping rather than projecting out-of-support observations would narrow the population; extending the support would require valid leisure and wage equations there; an observation or censoring model would retain the observations but add a new likelihood layer.

3 Household RURO model, assumptions and identification

3.1 Job space and structural choice probabilities

For a single adult, a working job is $j = (k, h, w)$, with occupation $k \in \{1, 2, 3, 4\}$, weekly hours $h \in [5, 70]$, and hourly wage w . Non-employment is one atom, denoted 0. The mixed reference measure is

$$\nu = \delta_0 + \sum_{k=1}^4 \delta_k \otimes dh \otimes dw. \quad (3)$$

The International Standard Classification of Occupations (ISCO) one-digit groups are aggregated into four civilian groups: manual/agricultural, service/sales, clerical, and managerial/professional/technical. ISCO zero lies outside this recode; the current mode assignment of employed armed-forces cases is disclosed and remains under decision.

Let $u_i(j; \theta_P)$ be deterministic utility and $g_i(j; \theta_O)$ an unnormalised structural opportunity density. With iid type-I extreme-value utility shocks of unit scale,

$$a_i(j; \theta) = u_i(j; \theta_P) + \log g_i(j; \theta_O), \quad Z_i^a(\theta) = \int_{\mathcal{J}_i} e^{a_i(j; \theta)} d\nu(j), \quad (4)$$

$$p_i(j; \theta) = \frac{e^{a_i(j; \theta)}}{Z_i^a(\theta)}. \quad (5)$$

For continuous working alternatives p_i is a density relative to ν ; at the atom its value is a probability mass. Density height is not interval mass.

The unbounded historical wage model is not certified for singles with positive consumption curvature. Under eventual positive-power budget growth on a positive-measure set of hours, its utility-weighted normalizer diverges. Finite priced nodes cannot establish the budget at infinity. The initial corrective candidate is therefore the bounded support $\mathcal{W} = [2, 150]$ euros per hour, with upper-cap sensitivity at 120 and 175. If $g_i^W(w | k)$ is the historical lognormal density, the corrected density is

$$g_i^{W,B}(w | k) = \frac{g_i^W(w | k)}{D_i(k; \theta_W)} \mathbf{1}\{2 \leq w \leq w_{max}\}, \quad (6)$$

$$D_i(k; \theta_W) = \Phi\left(\frac{\log w_{max} - \mu_i(k)}{\sigma}\right) - \Phi\left(\frac{\log 2 - \mu_i(k)}{\sigma}\right). \quad (7)$$

The parameter- and occupation-dependent divisor is part of the economic density and its derivatives. It is not a numerical sampling adjustment. The support is a disclosed approximation for a trimmed population, not an identified maximum. Compact support yields finite normalizers conditional on budgets being bounded on compact earnings sets and utility remaining on its positive domain.

3.2 Utility and budgets for singles and couples

Define the Box–Cox map

$$\text{BC}(x; t) = \begin{cases} (x^t - 1)/t, & t \neq 0, \\ \log x, & t = 0. \end{cases} \quad (8)$$

For singles,

$$u_i(j) = \omega_i \text{BC}\left(\frac{80 - h(j)}{10}; \theta_{\ell, g}\right) + \text{BC}\left(\frac{C_i(j)}{1911.108057855561}; \theta_c\right), \quad (9)$$

$$\omega_i = \beta_{\ell 0, g} + \beta_{\ell a, g} a_i + \beta_{\ell a^2, g} a_i^2 + \mathbf{1}\{g = f\} \beta_{\ell n, g} n_i, \quad a_i = (A_i - \bar{A}_g)/10. \quad (10)$$

A_i is age in years and n_i counts all resident household members younger than twenty. The child coefficient enters additively in the leisure weight; it is not exponentiated. The leisure weight is not itself a marginal rate of substitution. The local consumption compensation for leisure depends on both marginal utilities and the normalization scales.

For a couple, $j = (j_m, j_f)$ and the regimes are neither working, man only, woman only, and both working. Utility is

$$u_i^C(j_m, j_f) = \omega_{im} \text{BC}(\ell_{im}; \theta_{\ell m}) + \omega_{if} \text{BC}(\ell_{if}; \theta_{\ell f}) + \log\left(\frac{C_i(j_m, j_f)}{3821.448012098882}\right). \quad (11)$$

There is no direct cross-leisure term: $\beta_{\ell\ell} = 0$ in the specification being reported. Spouses are nevertheless dependent through joint choice, shared consumption, income effects, and nonlinear taxes and transfers.

Table 1: Population: final single-adult and both-flexible couple samples. Units: utility arguments are dimensionless; money and time units are shown in the rows. Measure: model specification. Reference: current-implementation model, with the bounded wage correction pending. Status: structural comparison, not an empirical result.

Feature	Singles	Couples
Decision	non-work or one job	joint four-regime package
Consumption	decider-only aggregation historically; all-member correction pending	all resident members
Consumption curvature	estimated Box–Cox	logarithmic
Leisure	one term	spouse-specific additive terms
Direct cross-leisure	not applicable	fixed at zero
Opportunity factorization	employment, hours, occupation, wage	spouse product conditional on household covariates
Dependence in choice	one decision maker	joint budget and joint utility
Sampling law	mixed pseudo-random and scrambled-Halton historical panel	joint-regime first, then spouse job coordinates

3.3 Opportunity factors

Using the uncentred representation, the single-adult working kernel is

$$g_i(k, h, w) = \exp\{E_i + O_{gk} + H_g(h)\} g_i^W(w | k), \quad g_i(0) = 1. \quad (12)$$

E_i contains the employment intercept, region and urbanization variables, and $\beta_{E,gsur}(10\ gsur_i)$. Thus a coefficient is applied to ten times the group unemployment rate; its exponentiated effect is a relative density factor, not a percentage-point employment effect. $O_{g1} = 0$ normalizes the first occupation group. The hours index has disjoint continuous bands, including $[33.5, 36.5)$ around thirty-five hours. It changes density height across an interval; it is not an atom at thirty-five hours. For example, the integrated mass of band b is its width times e^{β_b} divided by the hours normalizer.

For couples the uncentred opportunity kernel factors as $g_i^C(j_m, j_f) = g_{im}(j_m)g_{if}(j_f)$. This conditional factorization does not imply independent decisions, because equation (11) and the joint budget enter the same choice probability.

Identification. Preferences and opportunities enter a_i additively. Their separation relies on functional form, observed variation, and exclusion restrictions: local group unemployment, region and urbanization are assigned to access and excluded from preferences; demographic shifters enter declared preference or wage channels. The design does not nonparametrically recover capability sets, discrimination, amenities, or persistent unobserved heterogeneity. Sorting can invalidate a causal reading of geography. A benchmark comparison tests the consequences of a common density; it does not identify the preferred model.

Calculation, rationale, assumptions, alternative. The model calculates the probability density in equation (5) from utility and availability. It is used to avoid forcing restricted choices into tastes. It requires a proper mixed support, finite normalizers, valid channel exclusions, and unit-scale extreme-value shocks. A common-menu random-utility model is a coherent benchmark but removes household-specific access. A richer nonparametric opportunity process would weaken functional-form assumptions but requires new identifying variation and much denser data.

4 Estimation and numerical implementation

4.1 Structural density, choice probability, and proposal density

The structural opportunity density g_i is economic. The choice density p_i combines g_i with utility. The numerical proposal q_i is chosen by the researcher to place simulated alternatives; it is not an opportunity density. For a stochastic single-adult draw,

$$q_i(j) = q_E(e) [q_O(k \mid dgn_i, educ3_i) q_H(h) q_W(w \mid i, k)]^e. \quad (13)$$

q_H is the exact marginal sum over overlapping mixture components. For couples the proposal first samples a joint participation regime and then samples the job coordinates of each working spouse. The current wage coordinate uses a scrambled Halton point set and is dependent within household.

4.2 Historical criterion and authorized correction

Let j_i^* be the observed package, x_{ir} stochastic alternatives, and $a_i = u_i + \log g_i$. The historically implemented criterion is

$$\ell_i^C(\theta) = a_i(j_i^*; \theta) - \log \left[e^{a_i(j_i^*; \theta)} + \sum_{r=1}^R \frac{e^{a_i(x_{ir}; \theta)}}{q_i(x_{ir})} \right]. \quad (14)$$

It combines a unit-weight chosen row with importance-weighted stochastic rows. The criterion audit establishes that this is a finite- R approximation to a direct simulated population log-density, carrying an additional chosen-row term of order R^{-1} . It is not the exact conditional

sampled-alternatives likelihood. The estimates reported below were obtained from equation (14); they are not relabelled as corrected.

The authorised successor regenerates iid joint alternatives. For a sampled multiset D_i and multiplicity n_y , iid factorization gives

$$\Pr(y \text{ chosen} \mid D_i) = \frac{n_y \exp\{a_i(y) - \log q_i(y)\}}{\sum_{z \in D_i}^{\text{slots}} \exp\{a_i(z) - \log q_i(z)\}}. \quad (15)$$

The chosen package receives the same $-\log q_i$ correction as every other candidate, and repeated atoms retain multiplicity. The conditional likelihood is exact for its stated sampling experiment. That statement does not imply an unbiased finite-sample maximizer, exact coverage, or no benefit from enlarging a sampled menu. The current scrambled-Halton wage panel is not iid and cannot be made iid by changing stored corrections. It must be regenerated. Whether pilot proposal parameters are independent of the observed-choice sample, or instead must be cross-fitted, remains unresolved.

For population prediction and welfare, the relevant numerical object is direct integration of the structural population kernel. A generic importance estimate on working draws is

$$\widehat{Z}_i^a = e^{a_i(0)} + \frac{1}{R} \sum_{r: x_{ir} \neq 0} \frac{e^{a_i(x_{ir})}}{q_i(x_{ir})}. \quad (16)$$

The chosen package is an evaluation point in the numerator and is absent from this denominator. The atom and working integral must be scaled coherently; the $1/R$ cannot be removed from only one limb.

4.3 Inference and weights

Household-cluster-robust inference aggregates one score per household, permits arbitrary dependence among that household's simulated rows, and assumes independence across households. For interior coordinates θ_I ,

$$s_i = \frac{\partial \ell_i}{\partial \theta_I}, \quad \widehat{\text{Var}}_{CR1}(\hat{\theta}_I) = \frac{G}{G - K_I} H^{-1} \left(\sum_{i=1}^G s_i s_i' \right) H^{-1}, \quad (17)$$

where G is the number of households, K_I the number of interior coordinates, and H the negative Hessian. The factor $G/(G - K_I)$ is the adopted CR1 finite-sample adjustment. Active-bound coordinates require constrained or one-sided treatment; ordinary symmetric intervals are not reported for them. The current singles fit has 39 interior and 2 active-bound coordinates; the couples fit has 46 free, 45 interior, and 1 active-bound coordinates.

Survey weights make descriptive, fit, and welfare aggregates represent the target population. They do not enter the unweighted estimation objective. Conditional parameter uncertainty propagates equation (17) while holding the data, model, active set, policy build, numerical integration design, and welfare reference fixed. It is not numerical integration error and it is not normative-reference sensitivity.

Calculation, rationale, assumptions, alternative. Estimation calculates parameters from conditional sampled-menu probabilities; population prediction calculates equation (16). Separating these operations keeps one economic model while respecting two numerical questions. The conditional estimator requires iid joint draws, positive proposal support, correct multiplicities, and an independent or properly treated pilot proposal. Keeping the current dependent point set licenses direct integration, not the iid conditional factorization. A direct simulated maximum-likelihood estimator would retain the point set but has finite-integration bias and must treat the atom and normalization coherently.

5 Money-metric well-being and inequality decomposition

5.1 From opportunities to ex-ante value

For a coalition S of equalized factors, distinguish three objects:

$$G_{i,S} = \int_{\mathcal{J}} \tilde{g}_{i,S}(j) d\nu(j), \quad Z_{i,S} = \int_{\mathcal{J}} e^{u_{i,S}(j)} \tilde{g}_{i,S}(j) d\nu(j), \quad (18)$$

$$J_{i,S} = \frac{Z_{i,S}}{G_{i,S}} = \int_{\mathcal{J}} e^{u_{i,S}(j)} \hat{g}_{i,S}(j) d\nu(j), \quad \text{quad} V_{i,S} = \log J_{i,S}, \quad (19)$$

where $\hat{g}_{i,S} = \tilde{g}_{i,S}/G_{i,S}$. G is opportunity mass, Z is a utility-weighted normalizer, and J is their normalized ratio. No one symbol is used for all three. The calculation is ex ante: latent jobs are averaged before the logarithm and monetary inversion. Observed non-employment does not remove working opportunities from this integral.

Randomized quasi-Monte Carlo (RQMC) uses randomized, deliberately well-spread points for the latent-job integral. On scramble r with M common-support nodes,

$$\hat{J}_{i,S,r} = \frac{1}{M} \sum_{m=1}^M \exp\{u_{i,S}(J_{irm}) + \log \hat{g}_{i,S}(J_{irm}) - \log q_r(J_{irm})\}, \quad (20)$$

$$\bar{J}_{i,S} = \frac{1}{K} \sum_{r=1}^K \hat{J}_{i,S,r}, \quad \hat{V}_{i,S} = \log \bar{J}_{i,S}. \quad (21)$$

Averaging occurs before the logarithm and before inversion. WINT-1 follows the executed call path and finds no observed row on this common support. Therefore the chosen-row contribution to equations (21) is exactly zero. The support must still match the corrected structural density; reusing priced nodes outside the new support requires zero target contribution with the original proposal weight, not division by a random retained-node count.

5.2 Monetary inversion

Remove the consumption term from attained utility,

$$L_{i,S}(j) = u_{i,S}(j) - \text{BC}(C_{i,S}(j)/\lambda_c; \theta_{c,S}), \quad (22)$$

and define the reference map

$$\Phi_{i,S}(m) = \log \int_{\mathcal{J}} \exp\{L_{i,S}(j) + \text{BC}(m/\lambda_c; \theta_{c,S})\} \hat{g}_{i,S}(j) d\nu(j). \quad (23)$$

Money-metric well-being $W_{i,S}$ is the unique monthly euro amount satisfying

$$\Phi_{i,S}(W_{i,S}) = V_{i,S}. \quad (24)$$

The same flat disposable-consumption amount is inserted at every reference alternative, including non-employment. It replaces consumption in the comparator; it does not replace hourly wages, rerun EUROMOD, or remove the effect of actual earnings on attained value. Both sides use the same coalition's preferences, opportunity density, and non-consumption utility. Existence and uniqueness require a continuous, strictly increasing reference map over the permitted monetary bracket and finite positive normalizers. A one-hundred-euro difference means a one-hundred-euro monthly difference in the flat reference consumption needed to match attained ex-ante value; it is not necessarily a cash-transfer equivalent under the actual nonlinear budget.

5.3 Interpersonal comparison, equivalization, and inequality

The welfare unit is the modelled household. Survey weighting also occurs at the household level. Couples are unitary decision makers in this application; the model does not identify within-household allocation. The modified-OECD scale is

$$m_i^{OECD} = 1 + 0.5(N_{i,14+} - 1) + 0.3N_{i,<14}, \quad (25)$$

and equivalized welfare is $W_{i,S}/m_{i,S}^{OECD}$. The scale moves with the composition factor so the counterfactual state is internally coherent. Closure is a consequence of this choice, not its ethical justification. Raw household, equivalized household, and pooled individual comparisons answer different questions and are not forced to coincide with type-specific constants.

With normalized household survey weights ω_i , the implemented Gini is

$$I(S) = \frac{\sum_i \sum_j \omega_i \omega_j |W_{i,S} - W_{j,S}|}{2 \sum_i \omega_i W_{i,S}}, \quad \sum_i \omega_i = 1. \quad (26)$$

An income Gini and a money-metric well-being Gini differ because the latter includes preferences, leisure, and opportunity distributions. Atkinson or generalized-entropy indices would make inequality aversion or subgroup decomposability more explicit; they would change the social summary, not the underlying household welfare values. No uncomputed index is reported as a result.

5.4 Factor operators and grouped attribution

The factors are P (preferences), A (employment, hours, and occupation access), B (wage-offer or earning opportunities), and D (resources and composition/needs). Each operator substitutes declared reference primitives before normalizing the opportunity kernel and recomputing budgets, welfare, equivalence scales, and inequality. q and normalizers are numerical or derived objects, not separate economic factors. The corrected four-field resource classification and the treatment of job-history counters must be resolved before the D states are repriced.

For the first-level two-player table, let one mean equalized and let I^{ab} be inequality with preference status a and environment status b . One-factor equalization effects are $I^{00} - I^{10}$ and $I^{00} - I^{01}$. They are not Shapley contributions. The two-player Shapley allocation is

$$C_P = \frac{1}{2}[(I^{00} - I^{10}) + (I^{01} - I^{11})], \quad C_E = \frac{1}{2}[(I^{00} - I^{01}) + (I^{10} - I^{11})]. \quad (27)$$

It averages each marginal effect over both positions in an ordering and thereby allocates the interaction.

Inside $E = \{A, B, D\}$, the executed grouped rule is

$$C_i = \sum_{Q \in \{\emptyset, \{P\}\}} \frac{1}{2} \sum_{T \subseteq E \setminus \{i\}} \frac{|T|!(2 - |T|)!}{3!} [I(Q \cup T) - I(Q \cup T \cup \{i\})], \quad i \in E. \quad (28)$$

Contributions may be negative. Shares sum to the reducible difference between baseline and the fully equalized state; they sum to baseline inequality only if the fully equalized state has zero inequality. Neither closure nor the ordering rule identifies causal responsibility.

Calculation, rationale, assumptions, alternative. This stage calculates normalized ex-ante value, inverts it into monthly euros, measures its weighted inequality, and allocates the reducible difference across factors. It is used because chosen income alone omits leisure and opportunity variation while a raw utility index is not interpersonally comparable. It requires finite integrals, a monotone reference map, the unitary-household assumption for couples, a declared equivalence scale, and coalition operators that alter only their assigned primitives. Ex-post welfare, common-reference opportunities, compensating variation, another inequality index, or flat rather than grouped Shapley attribution would each answer a different normative question.

6 Behavioural and welfare evidence for singles and couples

6.1 Parameter interpretation and fit

The current coefficients are estimates of equation (14). The female singles child coefficient is an additive change in ω_i , not an exponentiated multiplicative effect. The group-unemployment coefficient applies to 10 *gsur*_{*i*}. Hours-band coefficients compare density heights, and the narrow thirty-five-hour band is continuous. Mean desired hours do not establish an every-worker claim. The external wish-to-work-more variable lacks an established coding direction, so no directional robustness conclusion is drawn from it.

Fit is evaluated by population probabilities from the structural model, not by conditional sampled-menu probabilities. Singles and couples are both core: couple fit uses the four joint participation regimes and spouse-specific hours, occupation, and wage margins. A fit comparison is external benchmarking unless the identifying argument itself uses that source. Public aggregate wage and hours benchmarks do not turn the model’s structural exclusions into identified causal effects.

6.2 Provisional welfare evidence

Table 2: Population: final single-adult and both-flexible couple samples. Units: Gini points and percentage share where marked. Measure: model-implied raw household money-metric well-being and first-level grouped attribution. Reference: the current paper’s primary welfare reference. Status: current-implementation estimates under equation (14) and the historical unbounded wage density; corrected results pending.

Quantity	Singles	Couples
Baseline raw well-being Gini	0.1343	0.1447
Preference attribution	6.3%	0.0073 points
Non-preference attribution	93.7%	0.1374 points

The provisional evidence points toward a large role for non-preference circumstances in both populations. That qualitative pattern is useful for motivating the corrected analysis, but the corrected magnitude, factor ordering, and statistical comparison cannot yet be drawn. The nested access, earnings, resources, and composition percentages are not reproduced because the corrected field classification requires repricing. Couples are not relegated to future work: their joint model, sample, fit targets, welfare construction, and pending correction are part of the same empirical argument.

Actual non-workers are included in ex-ante integration over working and non-working alternatives. Their opportunities therefore affect welfare even when the chosen state is non-employment. Reference flattening applies only to the comparator in equation (23); actual earnings opportunities remain in equation (19).

6.3 Numerical, parameter, and reference uncertainty

For a fully recomputed scalar T , a leave-one-scramble-out jackknife omits one entire independent scramble, reruns averaging, inversion, equalization, Gini, and attribution, and measures integration variability. If there are K scrambles,

$$\bar{T}_{(-)} = K^{-1} \sum_r T_{(-r)}, \quad se_{jack} = \left[\frac{K-1}{K} \sum_r (T_{(-r)} - \bar{T}_{(-)})^2 \right]^{1/2}. \quad (29)$$

This is numerical integration uncertainty. Conditional parameter intervals propagate equation (17). Reference sensitivity changes a normative input to equations (23)–(24). The three are reported separately. No wider reference range is called a confidence interval, and no narrow RQMC band is evidence that parameter uncertainty is small.

7 Comparisons and sensitivity

7.1 Common-opportunity random-utility model

The common-opportunity comparison replaces every household-specific density by one normalized density,

$$\bar{g}(0) = \pi_0, \quad \bar{g}(1, k, h, w) = \pi_1 \bar{p}(k) p_H(h) \phi_{LN}(w \mid \bar{\mu}(k), \hat{\sigma}), \quad (30)$$

with $\pi_0 + \pi_1 = 1$ and each conditional component integrating or summing to one. It does not set g to zero, which would remove alternatives. Taste terms are re-estimated under the restriction. The recorded criterion difference is not a likelihood-ratio test because surviving opportunity coefficients were held at the RURO values rather than freely re-optimized. It is descriptive benchmark evidence, not formal identification.

7.2 Geography and external comparison

Geographic attribution changes the location-related limb of current employment access. It is therefore conditional on the model’s channel assignment. It is not a lower bound on the total importance of place: omitted wage, hours, amenity, migration, and sorting channels may offset or reinforce it. Comparisons with other countries or papers require matching the population, welfare unit, numerator, denominator, policy question, and reference environment. Percentages from reform compensating variation are not directly comparable to shares of cross-sectional well-being inequality.

8 Limitations and conclusion

The first limitation is computationally immediate: the corrected estimator, wage support, and downstream results do not yet exist. Second, bounded wage support is a disclosed structural approximation, not an identified supremum; its cap sensitivity and budget coverage must be reported. Third, the target population excludes same-sex couples, older or studying decision makers, households with additional employable adults, and workers outside the retained support. Fourth, geography is not causal. Fifth, opportunity factors are parametric densities rather than literal choice sets. Sixth, the couple model is unitary and fixes direct cross-leisure interaction to zero. Seventh, the one-euro consumption floor and seeded take-up traits are modelling assumptions. Eighth, the resource/composition partition still requires decisions on job-history counters and non-cash benefits. Ninth, the current data do not establish persistent unobserved heterogeneity merely because attempted extensions were weakly recovered.

The paper’s substantive claim is therefore deliberately narrower than a final percentage. Unequal employment, hours, occupation, and wage opportunities are economically distinct from preferences and can matter for ex-ante well-being even when chosen outcomes look similar. Resources and household composition are separate non-preference channels needed for exhaustive accounting. The current implementation suggests these distinctions matter for singles and couples, but corrected magnitudes await the authorized refit and downstream regeneration.

Similar estimated preferences, different employment-access environments

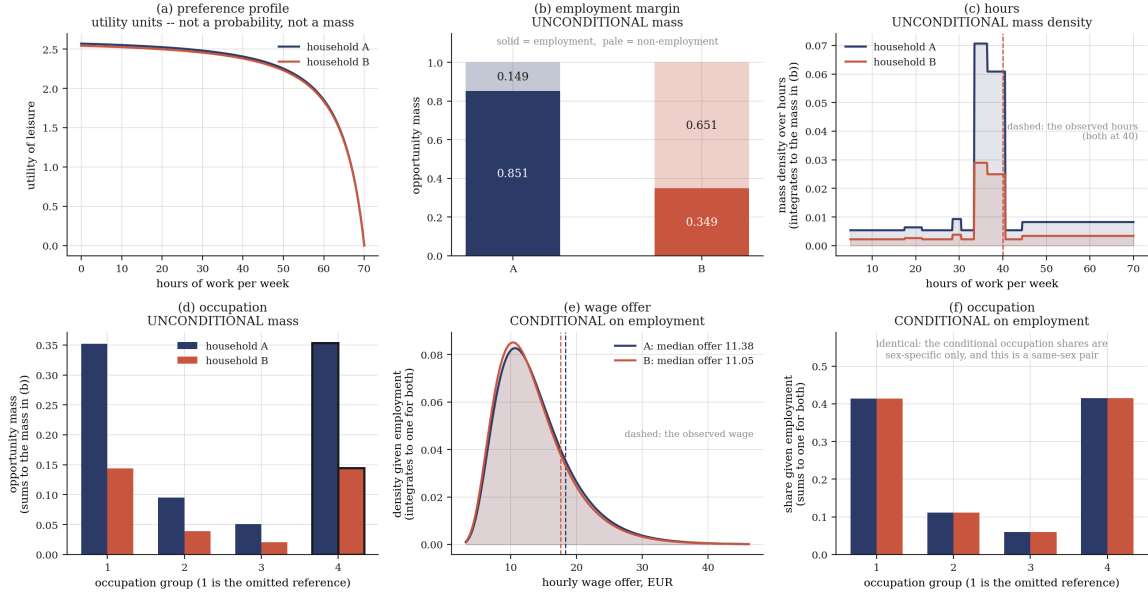


Figure 4: **Similar estimated preferences, different employment-access environments.**

The two single men are selected by the stated matching rule among employed households in the same observed occupation group, hours band and wage quintile. Their estimated leisure-preference curves are close, not identical. Panel (a) plots the leisure component of utility; panels (b) and (c) show the specified opportunity masses over hours and occupations; panel (d) shows conditional wage-offer densities and marks the observed wages. In this same-sex comparison the hours and occupational profiles are common conditional on employment, so their unconditional mass differences are driven primarily by the employment-access margin. The example illustrates a model-conditional contrast, not a causal regional effect, an individual ability ranking or a recovered count of available jobs. Population: two anonymized employed single men in occupation group 4, a common hours band, and observed-wage quintile 4. Units: weekly hours and euros per hour. Measure: utility component and opportunity mass/density, not choice probability. Reference: each household's estimated profile under the current implementation. Status: model-implied diagnostic; corrected estimator and support results pending.

Note: The forward rule restricts preference distance to the lowest decile of admissible distances and maximizes total-variation opportunity distance, subject to the displayed clean-separation threshold. The pair is matched on occupation group, hours band and wage quintile; neither preferences nor observed job bundles are asserted to be identical.

A Reproducibility map

The canonical research notebook is `MNL/experiments/JMP_SEMINAR_SPRINT/JMP_research_lab.ipynb`. It exposes authenticated frames and cached draws, replays the current criterion, loads inference and fit outputs, and dispatches cached welfare and figure stages. Raw survey construction, arbitrary fresh alternative generation, EUROMOD repricing, and external benchmark construction remain external stages. Engine capability, the package actually selected by the notebook, and device evidence are separate facts. The local parity record compares Torch on CPU with JAX for the final singles model and reports CUDA unavailable on that machine. A CUDA profile exposed by the engine is therefore not a measured local CUDA result, and no singles result establishes couples device support.

The corrective sources are `MNL/docs/corr/target_model_and_integrability_v1.md`, `MNL/docs/jmp_methodology/JMP_sampled_alternatives_criterion_audit_v1.md`, `MNL/docs/corr/fr2016_source_to_estimation_sample_audit_v1.md`, and `Job_Market_paper/reports/model_extraction_v1.md`. The source audit’s field dictionary and machine-readable evidence supply units, policy identifiers, and sample counts. Empirical numerals in this manuscript are generated from `reports/numbers_of_record_v1.json` and the regenerated final descriptives registry by `manuscript/build_paper_v2_numbers.py`.

B Pending substantive calculations and decisions

1. Generate iid joint alternatives; establish independent or cross-fitted pilot proposals; implement equation (15); and re-estimate singles and couples.
2. Implement equations (7) in estimation, derivatives, prediction, and welfare; price uncovered nodes; and run the stated upper-cap sensitivity. The old unbounded singles normalizer is not unconditionally certified because finite pricing does not establish budget growth at infinity.
3. Aggregate all resident-member disposable resources for singles, retain the full accounting outputs, and re-estimate.
4. Decide the observation model for employed hours outside the support and the treatment of employed ISCO-zero decision makers.
5. Finalize economic factor allocation for job-history counters and non-cash benefits; then reprice the corrected resource/composition states. Until then the nested percentages cannot be interpreted.
6. Recompute parameters, population fit, welfare, uncertainty, and grouped attribution for both household types. No corrected component ranking is yet available.

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